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Prof. Patrick Siarry
Editor-in-Chief
Engineering Applications of Artificial Intelligence

Dear Prof. Siarry,

We are pleased to submit our manuscript titled **“Learning-based optimisation framework for flexibility aggregators in ancillary service markets”** for consideration in the *Engineering Applications of Artificial Intelligence*.

This work addresses the challenge of optimizing offering strategies for flexibility aggregators operating in the European manual frequency restoration reserve market. Our research proposes a constraint learning-based optimization framework that integrates machine learning surrogate models into optimization problems. Specifically, we introduce a novel linear programming (LP) reformulation for rectified linear unit (ReLU) deep neural networks (DNNs), enabling scalable and accurate modelling of prosumers' price responsiveness and flexibility behaviour.

Key contributions of this paper include:

- Proposal of a method to “convexify” ReLU DNNs, enabling their integration as tight LP reformulations within optimisation problems.
- Development of an aggregator's profit maximisation model incorporating a nonlinear, nonconvex prosumer responsiveness function.
- Application of the optimisation with constraint learning framework by learning the prosumer responsiveness function using a convexified ReLU DNN and embedding it as an LP reformulation within the aggregator's profit maximisation problem.
- Comprehensive comparison of the proposed reformulation against state-of-the-art ReLU DNN reformulations and the conventional piecewise linearisation, evaluated across a range of optimisation parameters and ReLU DNN architectures.

We believe this manuscript is well aligned with the journal's focus on the application of artificial intelligence to make engineering problems tractable, particularly in leveraging ReLU DNNs for optimisation in energy market contexts.

We confirm that the manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and consent to its submission to *Engineering Applications of Artificial Intelligence*.

Thank you for considering our submission. We look forward to your feedback.

Sincerely,
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